Spatial Analysis of Earthquakes

Introduction

This report presents a simple earthquake analysis project using QGIS, a Geographic Information System (GIS) tool. The project maps earthquake events recorded between **1st September to 8th September 2025**, highlighting their magnitude, distribution, and proximity to major tectonic boundaries. Supplementary layers such as world borders and major cities are included to provide geographic context.

Objective

The objective of this report is to demonstrate how **spatial data** can be combined with freely available resources (USGS earthquake records, Natural Earth map layers, and tectonic plate boundary data sets) to visualise earthquake patterns and show the relationship between recent seismic activity and tectonic plate structures.

Methodology

The analysis was carried out in several stages, beginning with data collection and followed by data integration, visualisation, and map design.

1. **Data Collection**

   - **Earthquake Data (Point Features)**:
     A CSV dataset of global earthquakes recorded between *1st September and 8th September 2025* was obtained from the **United States Geological Survey (USGS)**. The dataset included **key attributes** such as latitude, longitude, magnitude, and depth.

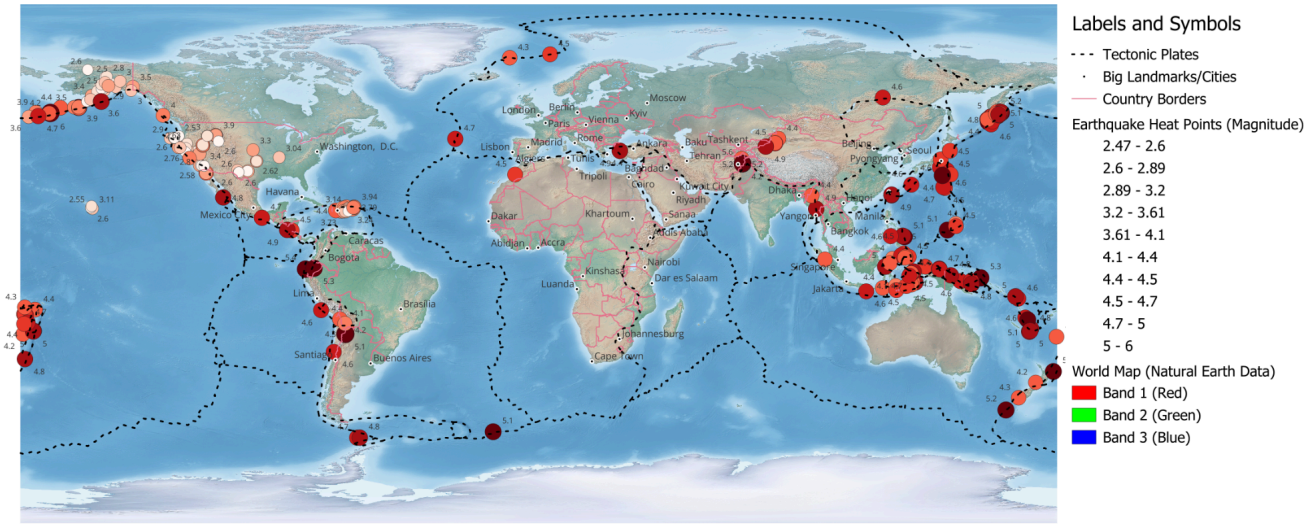
   - **Base Map Layers (Raster Imagery)**:
     Physical and cultural map data were taken from the **Natural Earth** dataset, including country borders and major cities for geographic context.

   - **Tectonic Plate Boundaries (Polygon Layer)**:
     Global tectonic boundary shapefiles were downloaded from an **open-source** repository *(Fraxen/Peter Bird PB2002)* and used to visualise the relationship between earthquake activity and plate structures.

2. **Symbology and Visualisation**

   Earthquakes were symbolised according to their **magnitude** using **graduated circle sizes**, with larger magnitudes represented by larger points. **Tectonic plate boundaries** were styled as **bold black lines** to contrast with the base map and highlight their connection to earthquake clustering. Major world capitals were included as reference points but filtered to avoid overcrowding.

Analysis



As shown in the final map, the majority of earthquakes during the study period occurred along tectonic plate boundaries, supporting the well-established link between seismic activity and plate tectonics.

Along the **western margin of North America**, clusters of small-magnitude earthquakes (<4.0) were observed. These lighter events, represented by white and pale orange symbols, are typical of the tectonic stress along the **San Andreas Fault system** and the broader Pacific Plate boundary region.

In contrast, the **Asian-Oceanic plate boundary region** displayed a concentration of stronger earthquakes, ranging from 4.0 to 6.0 in magnitude. These events occurred particularly around the **Philippine Sea Plate** and the **Indonesia-Papua New Guinea region**, where complex subduction zones are known for producing higher-magnitude seismic events.

The **Pacific "Ring of Fire"** overall showed consistent activity, including smaller scattered events across Japan, Papua New Guinea, and the western coastlines of South America. This aligns with its status as the most seismically active region on Earth.

Interestingly, a small number of earthquakes occurred away from the major plate boundaries. These **intraplate earthquakes** are less common but highlight that seismic activity can also occur within continental interiors due to ancient fault systems or localised stress release.

In conclusion, magnitude distribution during this period indicates that the largest earthquakes were clustered around subduction zones in Asian and the western Pacific, while transform boundaries (e.g., San Andreas) primarily experienced lower-magnitude tremors.